



Research Paper

Quantifying transfer coefficients in battery recycling routes for material flow analysis: current state and future scenarios

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ABSTRACT

The growing volume of end-of-life batteries in the European Union necessitates robust quantification of secondary raw material recovery potential to support material flow analysis and policy evaluation. However, empirical data on recycling process performance remain limited. Transfer coefficients (TCs) quantify the fraction of a material transferred from input into a specific output stream at each process step. The product of step-wise TCs along a route determines recovery for each element and links process-level performance to system-level recovery outcomes. This study compiles route-specific TCs for five lithium-ion battery recycling routes encompassing pyrometallurgical, hydrometallurgical, mechanical, and direct recycling, alongside established processes for zinc-based, nickel-based, and lead-acid batteries. TCs compiled from patents, peer-reviewed literature, and industry data were integrated into a Python-based recovery model, generating 36,358 scenario-ready values from 2010 to 2050. Two scenarios are analysed: business-as-usual, with technological stagnation and continued black mass export, and recovery scenario aligned with EU Battery Regulation targets. A gap analysis reveals highly element-specific challenges. While TCs for cobalt, nickel, and copper already approach regulatory thresholds, lithium recovery from pyrometallurgy-based routes are far below the regulatory target. Non-regulated materials such as graphite, manganese, and phosphorus show zero or marginal recovery, a structural blind spot in the regulatory framework. A structural decomposition shows that process-level efficiency is the dominant driver of system-wide recovery for most elements, whereas route composition acts as a secondary and often counteractive effect. The publicly available TC dataset provides a transparent basis for secondary supply estimation while highlighting uncertainties requiring industrial-scale validation and chemistry-differentiated parameterisation.

1. Introduction

Global battery demand is rising rapidly across all major chemistries, with lithium-ion batteries (LIBs) driving most of this growth. According to the Urban Mine Platform, approximately 3.3 Mt of batteries were placed on the market in the EU in 2025, with projections indicating an increase to 8.4 Mt by 2050 (Maisel et al., 2026b; UrbanMinePlatform 2026). While lead-acid batteries remain the largest segment by mass and are supported by a well-established recycling system, LIB volumes are growing fastest and nickel- and zinc-based chemistries represent smaller but persistent flows (Maisel et al., 2026b; UrbanMinePlatform 2026). The rapid growth in LIB stocks from electrified mobility and stationary storage is generating large-scale end-of-life (EoL) flows, making recycling an increasingly strategic lever to secure critical battery metal

supply and reduce import dependence (European Commission 2024; IEA 2025). Although recycling capacities are increasing significantly across the European Union (EU), processing and material recovery remain constrained by bottlenecks such as limited black mass refining capacity and continued exports for downstream treatment (Stephan 2025). A recent European assessment reveals that existing recycling capacity would be insufficient to process over 78 % of forecasted EoL LIBs, while achieving high recoveries for several transition metals such as copper (Cu), cobalt (Co), and nickel (Ni) (around 90 %) but much lower recoveries for lithium (Li) (42 %) and iron (Fe) (35 %) in current full-scale processes (Bruno and Fiore 2023).

The EU's import dependence on critical raw materials for battery manufacturing, combined with low recycling rates for lithium and natural graphite, further undermines domestic raw material security and

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limits circularity options (Matos et al., 2022). Currently, secondary materials from EoL LIBs contribute only a small share of the metals required for new LIB production, as recycling infrastructure remains insufficient and return volumes are low relative to future demand (Bruno and Fiore 2024). However, policy interventions in the EU are expected to alter this trajectory significantly. The EU's Battery Regulation (EU 2023/1542) and associated material recovery provisions set ambitious targets for collection, recycling efficiency, and material recovery for lithium and other battery metal, thereby enhancing the secondary supply of critical raw materials (European Commission, 2023).

1.1. Problem statement

Transfer coefficients (TCs) parameterise how mass is distributed among process stages and output flows in material flow analysis (MFA) and are therefore central to scenario modelling and policy evaluation. Accurate, route-specific TCs are essential because both recycling performance indicators and environmental outcomes are sensitive to process yields and configurations. However, the empirical literature reports recovery and process performance in heterogeneous, chemistry-specific ways that are often not directly comparable across studies, complicating the derivation of robust TCs for system-level analysis.

A growing body of literature addresses battery recycling performance, yet these studies remain fragmented in scope, resolution, and policy linkage. Comparative LCA and process evaluations such as (Velázquez-Martínez et al., 2019) or (Choi et al., 2025) characterise individual routes but rarely report harmonised, step-wise TCs transferable across studies. Sensitivity analyses such as (Wilke et al., 2024; Rinne et al., 2024) or (Ciez and Whitacre 2019) demonstrate the dependence of recovery on input composition but do not provide a multi-route, multi-chemistry parameter set. Review-type contributions such as (Wagner-Wenz et al., 2023; Wang et al., 2022) or (Salehi et al., 2023) map the diversity of process combinations without deriving quantitative mass balances, while MFA and urban-mine studies such as (Husmann et al., 2025) or (Song et al., 2019) operate at system level and inherit the same reporting inconsistencies. Chemistry-specific work such as (Dai et al., 2019) or (Zou et al., 2024) reveals interaction effects in mixed feeds but is confined to single process stages. No existing study combines route-specific TCs across both LIB and conventional battery chemistries with a forward-looking scenario dimension and an explicit mapping to the current EU regulatory framework (Regulation (EU) 2023/1542 and Delegated Regulation (EU) 2025/606), the gap addressed in this study.

1.2. Aim and scope of the study

The present study addresses these gaps by pursuing two complementary objectives: (a) quantifying how TCs vary across technical recycling routes and battery types, and (b) developing forward-looking recycling scenarios that modify material transfer patterns and improve the recoverability of critical elements. Unlike most existing studies, which focus exclusively on LIBs, this work encompasses the full spectrum of established battery technologies, including LIBs, zinc-based (Zn) batteries, nickel-cadmium (NiCd), nickel-metal hydride (NiMH), lead-acid (Pb) batteries and lithium primary batteries, thereby enabling a cross-technology comparison of recycling performance and material recovery. The study makes four main contributions: (i) a compiled, route-specific TC set covering LIB and conventional chemistries, (ii) a publicly available dataset of 36,358 scenario-ready values (2010–2050), (iii) a Python-based recovery model and (iv) a gap analysis against Regulation (EU) 2023/1542 and 2025/606, bridging process-level evidence and system-level secondary-supply estimation.

1.3. Regulatory framework

Material recovery from waste batteries in the EU is governed by Regulation (EU) 2023/1542 (the Battery Regulation) and its supplementing Commission Delegated Regulation (EU) 2025/606. Part C of Annex XII sets minimum recovery targets for cobalt, copper, lithium, nickel, and lead: for lithium, 50 % by 31 December 2027 and 80 % by 31 December 2031; for cobalt, copper, and nickel, 90 % and 95 % for the same deadlines (European Commission 2023). As recovery is reported annually, the first full reporting years after these deadlines are 2028 and 2032, used as scenario reference years in this study (Section 2.5).

Commission Delegated Regulation (EU) 2025/606 establishes the methodology for calculating and verifying these rates and defines how system boundaries and recovery must be treated. Three provisions are decisive. First, material recovery is credited only at the calculation point for the recovery of material (RM calculation point), i.e. where a target material is recovered in a form that can substitute primary material in manufacturing (Annex, Section 3(2)) (European Commission 2025). Second, black mass is classified as an intermediate fraction that “cannot be considered as output fraction” (Section 1(5)); black mass exported for downstream treatment therefore does not, in itself, count as recovered material (European Commission 2025). Third, lithium contained in slag “is not taken into account in calculating the rate of recovery of materials” (Section 1(4)) (European Commission, 2025).

These provisions frame the interpretation of the transfer coefficients and scenario results derived in this study. Where the model reports exported black mass or slag-borne lithium as recovered, such values represent technical recovery potential rather than rates creditable under Regulation (EU) 2025/606; this distinction is made explicit in the results and discussion (Sections 3 and 4).

2. Methodology

The methodology of this study follows a sequential workflow that links process-level recycling evidence to system-level secondary-material recovery. An overview of the overall research procedure, from TC derivation to scenario modelling and system-level recovery evaluation, is provided in Fig. 1.

2.1. Literature review and derivation of transfer coefficients

To derive TCs, an extensive literature review of battery recycling covered the battery types Zn-based, Pb batteries, lithium primary batteries, NiCd, NiMH, and LIBs. Studies were included if they reported quantitative mass-balance data or element-specific recovery rates, while publications lacking process-level mass-flow information were excluded. Both peer-reviewed articles and grey literature were considered. Each reference was classified by its scale of origin lab-scale (L), pilot-scale (P), industrial-scale (I), or review paper (R), as shown in Table 1. Details of the database search and the screening procedure are provided in the Supplementary Information.

Data for battery TCs were collected using a common framework within a TC template (Yamamoto et al., 2026). The harmonised code lists are publicly available (Kippert et al., 2026), promoting transparency, interoperability, and consistency across datasets. Given the scarcity of quantitative mass-balance data, no statistical synthesis across sources was performed. For each process step, the single most representative available source was adopted, prioritising industrial-scale data. Review papers were used for process screening and source identification, not as primary TC sources. Where no dedicated source existed, values were extrapolated from related process steps (see Sections 4.2.1 and 4.2.2). These TCs were projected forward, accounting for anticipated technological developments and scenario-specific assumptions (Section 2.5), including back-casting procedures to ensure coherence with the defined scenario narratives.

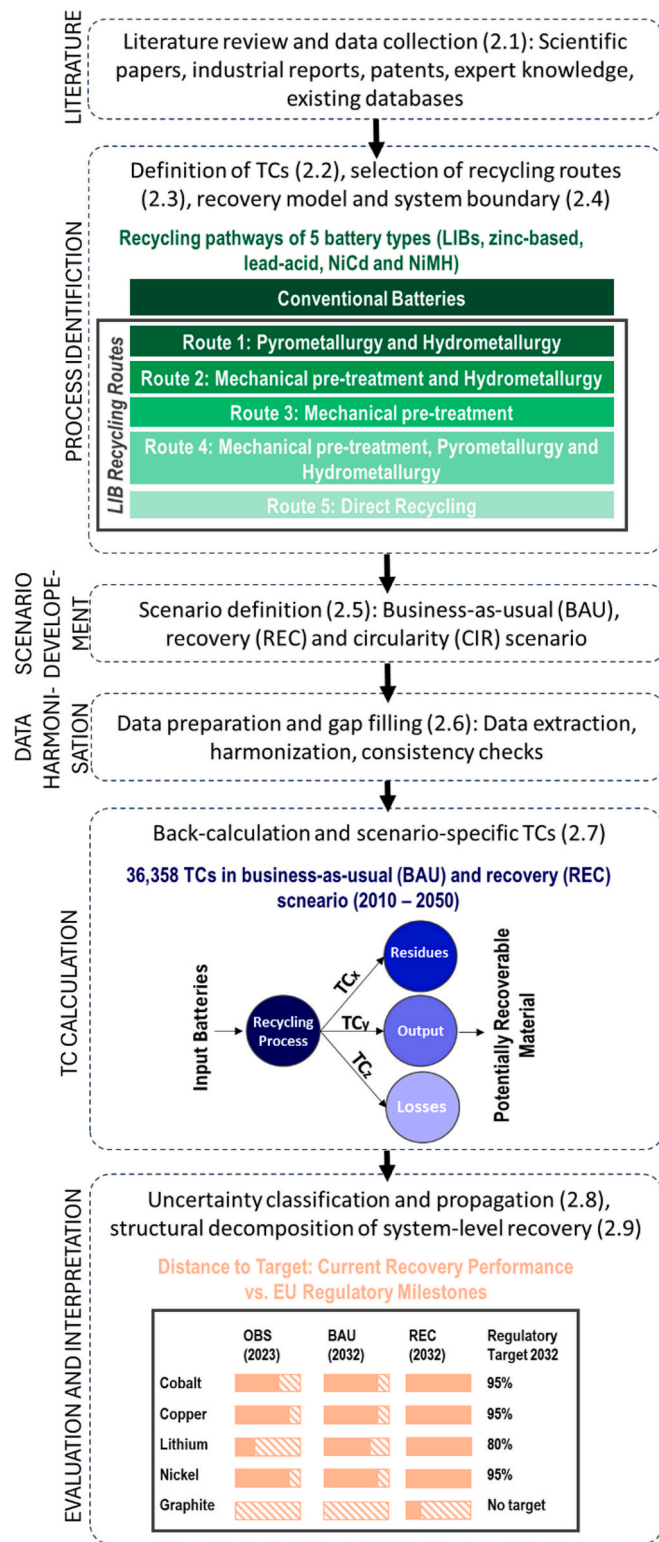


Fig. 1. Overview of the research procedure.

2.2. Definition and calculation of transfer coefficients

TCs are dimensionless parameters expressing the fraction of a material transferred from an input to a specific output stream during a defined unit operation or process (Brunner and Rechberger 2004). In the context of battery recycling, TCs quantify how much of each material is recovered into valuable output fractions versus being lost to process

residues or secondary streams such as slag (Vaccari et al., 2024). The TC for material i at process step j is defined by Equation (1), where $m_{i,input}$ denotes the mass of material i in the input and $m_{i,j,output}$ denotes the mass of material i in the corresponding output:

$$TC_{ij} = \frac{m_{i,j,output}}{m_{i,input}} \quad (1)$$

The material i represented by a TC can refer to elements, materials, components, or product layers. The TC ranges from 0 (complete loss) to 1 (complete recovery). This mass-based definition is consistent with the material recovery rate r_{RM} defined in Regulation (EU) 2025/606, which expresses recovery as the ratio of the mass of a target material in the output fraction to its mass in the input fraction (European Commission, 2025). For a multi-output recycling system comprising streams such as product-fractions, slag, residues, or losses, the sum of all TCs for material i must equal unity, under mass balance closure as shown in Equation (2):

$$\sum_j TC_{ij} = 1 \quad (2)$$

TCs thus serve as the fundamental building blocks for constructing mass balances in MFA and for assessing environmental and economic performance of recycling systems, and are specific to a given technology (Brunner and Rechberger 2004).

2.3. Identification and Selection of recycling routes

The available recycling technologies and their process steps were examined for Zn-based batteries, NiCd, NiMH, Pb batteries, and LIBs, including lithium primary batteries. Each battery chemistry has distinct recycling pathways, varying in technical maturity, recovery efficiency, and process configuration. For sodium-ion batteries (SIBs), only mechanical pre-treatment is assumed.

For LIBs, there is no single, standardized process, but rather, a range of process combinations is applied, depending on the operator, available infrastructure, and technological capabilities. The primary unit operations include discharging, mechanical treatment, thermal treatment (pyrolysis), pyrometallurgy, and hydrometallurgy. The discharging step is excluded from this study, as no material transfer occurs. Based on the literature review (Section 2.2), covering both industrial and developing processes and their recycling capacities, five representative routes were identified: Route 1: Pyrometallurgy combined with hydrometallurgy; Route 2: Pyrolysis (optional), mechanical pre-treatment, and hydrometallurgy; Route 3: Mechanical treatment only; Route 4: Pyrolysis (optional), mechanical pre-treatment, pyrometallurgy, and hydrometallurgy as well as Route 5: Direct recycling (LFP only).

2.4. Recovery model structure and system boundary

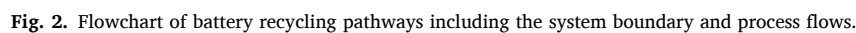
To operationalise the TC dataset, a Python-based recovery model was developed tracing material flows from collection through all recycling process stages to secondary raw material output. The Python model used to generate and apply the transfer coefficients, including all scripts, configuration files, and documentation, is available at GitHub (Yamamoto and Maisel, 2026). The model structure and comprehensive flowchart of all battery types is visualised in Fig. 2. Inflows enter the system at the collection stage, pass through chemistry-specific sorting nodes, and proceed along the different recycling routes (Section 2.3). Outflows crossing the system boundary represent secondary raw materials potentially recovered as well as material losses to residues or exports. Each mass stream is assigned a unique code name (see Supplementary Information), linking a specific flow to its corresponding TC value. The recovery model computes, for each combination of battery type, route, scenario, and reference year, the TCs along the respective flow path, providing the computational basis for the scenario results presented in Section 3.

Table 1

Literature on TCs and recycling processes for the 5 battery types including the primary research focus.

Reference	Zinc-based	Primary lithium	Lead-acid	NiCd	NiMH	LIB	Primary research focus	TCs	Recycling Process	Scale of Origin
(Salgado 2003)	■						Hydrometallurgy of zinc batteries		■	L
(Espinosa and Tenório 2004)				■			Vacuum distillation for NiCd batteries	■		L
(Ali et al., 2004)				■			Vacuum distillation for NiCd batteries	■		L
(Salomone et al., 2005)			■				Environmental Assessment	■	■	I
(Fisher et al., 2006)	■	■	■	■	■	■	Life Cycle Assessment		■	I
(Rombach et al., 2006)	■						Calculation of mass balance, Waelz kiln process	■	■	I
(Müller and Friedrich 2006)					■		Recycling process for NiMH		■	P
(UBA 2007)			■				Recycling of hazardous wastes		■	I
(Ferella et al., 2008)	■						Mechanical and hydrometallurgy	■	■	L
(Sayilgan et al., 2009)	■						Review of recovery technologies		■	R
(Briffaerts et al., 2009)	■						Waste battery treatment options	■	■	I
(Baba et al., 2009)	■						Pyro- and hydrometallurgy of zinc batteries	■	■	L
(Errington et al., 2010)			■				ISAMELT for Lead Recycling	■	■	I
(Granata et al., 2012)		■			■	■	Recycling Processes	■	■	P
(Xará et al., 2014)	■						Life Cycle Assessment		■	I
(Zhan and Xu 2014)				■			E-Waste Recycling	■	■	R
(Cucchiella et al., 2014)	■						Life Cycle Assessment	■	■	I
(Hanisch et al., 2015)						■	Recycling of LIB		■	R
(Nowak et al., 2015)						■	Extraction of LIB electrolyte	■		L
(Deep et al., 2016)	■						Hydrometallurgy	■	■	L
(JRC 2017)	■	■	■	■	■	■	Best Available Technique Recycling		■	I
(Wu et al., 2017)			■				Economic Assessment	■	■	I
(Petranikova et al., 2017)					■		Hydrometallurgy for NiMH batteries	■		P
(Ebin et al., 2018)					■		Recycling of NiMH batteries	■	■	P
(Ballantyne et al., 2018)			■				Lead-acid battery recycling		■	R
(ACCUREC 2018)						■	Battery Recycling Datasheet	■	■	I
(Kwade and Diekmann 2018)						■	Recycling of LIBs	■	■	R
(Tran et al., 2018)	■						Life Cycle Assessment		■	I
(Pan et al., 2019)			■				Sustainability evaluation		■	I
(Chen et al., 2019)						■	Recycling of EoL LIBs	■		R
(Liu et al., 2019)					■	■	Recovery from NiMH and LIBs	■	■	L
(Sommerfeld et al., 2020)						■	Pyro- and Hydrometallurgy for LIBs	■		L
(Lundström et al., 2020)	■						Recovery of high-purity silver	■	■	L
(Calin et al., 2021)	■						Electrostatic technology		■	P
(Mukhlis et al., 2021)	■						Technoeconomic study	■	■	L
(Blumbergs et al., 2021)				■			Cadmium Recovery from NiCd	■	■	R
(Pinheiro et al., 2021)					■		Mechanical Recycling of NiMH	■	■	L
(Farhad et al., 2022)	■	■	■	■	■	■	Recycling Processes		■	R
(Liu et al., 2022)						■	LIB Recycling	■	■	L
(Latini et al., 2022)						■	LIB Recycling	■	■	I/R
(Wagner-Wenz et al., 2023)						■	LIB Recycling Routes		■	R
(Wang et al., 2022)						■	LIB Recycling Methods, review	■	■	R
(Salehi et al., 2023)					■		Recovery of REE from NiMH	■	■	R
(Bhar et al., 2023)						■	LIB Recycling		■	R
(Dobó et al., 2023)						■	LIB Recycling, review	■	■	R
(Li et al., 2023)						■	Recovery from LIBs, lab scale	■		L
(Xu et al., 2023)						■	Pre-treatment of LIBs, review		■	R
(Nickelhütte Aue 2023)						■	Recycling Efficiencies of LIBs	■		I
(Hu et al., 2024)						■	Phosphorous Recovery from LFP	■		L
(Jiang et al., 2024)						■	Phosphorous Recovery from LFP	■		L
(Roy et al., 2024)						■	Direct Recycling of LIB	■	■	R/I
(Valdivia Huaman 2024)			■				Life Cycle and Material Flow Analysis	■	■	I
(Zhao et al., 2024)			■				Review of recovery technologies		■	R
(Forte et al., 2024)						■	Recycling of Co from LIBs	■	■	I
(Hämmer and Wambach 2024)						■	Recyclability of Smartphone Case Study	■	■	I
(Vaccari et al., 2024)						■	Performance of LIB Recycling	■	■	I
(Wilke et al., 2024)						■	Mechanical Recycling of LIB	■	■	P
(Woeste et al., 2024)						■	Techno-economic Assessment of LIB Recycling	■	■	I
(Zanoletti et al., 2024)						■	LIB Recycling		■	R
(Nickelhütte Aue 2024)						■	Flowchart of Battery Recycling		■	I

Lab scale (L), pilot scale (P), industrial scale (I), Review paper (R).



2.5. Scenario definitions

The BAU scenario assumes no major changes in battery waste management practices beyond current trends. While electrification leads to increasing end-of-life battery volumes, recovery performance remains largely unchanged due to the absence of significant technological advances and additional policy incentives (Islam et al., 2025; IEA 2025). This scenario serves as the baseline against which improvement requirements are quantified. In the REC scenario, regulatory targets are met at the dates specified by the Battery Regulation (Section 1.3) (European Commission, 2023), supported by technological innovation, enhanced collection systems, and a transition towards a circular battery

value chain with industrial-scale recycling and reduced reliance on primary critical raw materials, in line with the European SET Plan vision for a circular battery value chain (European Commission 2026). This target-compliant pathway identifies the required process-level TCs. TCs for NiMH, NiCd, Pb- and Zn-based batteries are assumed to remain constant, as established processes already operate at near-maximum efficiency. For LIB recycling pathways, the EU Battery Regulation targets (Section 1.3) serve as guiding benchmarks. Under BAU, the recovery levels required by the regulatory 2027 deadline are adopted as upper bounds for 2032 within the modelling, while under REC, targets are met at the specified dates and remain constant thereafter. In the CIR scenario, TCs remain unchanged relative to REC scenario; instead, it explores an increased share of batteries diverted to second-life applications prior to recycling (Helander and Ljunggren 2023). This scenario isolates the effect of lifetime extension on secondary material availability and timing of EoL flows. As it does not affect the TCs and only influences an MFA-related parameter, it is not described further.

2.6. Data preparation and gap filling

Prior to analysis, the raw TC data underwent several cleaning and harmonisation steps by removing duplicates and consolidating records prone to double counting (e.g. “ferrous metals” and “Fe” for the same flow) and all identifiers were validated against the master list of stocks and flows. Complete annual time series were obtained via a three-step interpolation procedure: zero before the first reported year, linear interpolation for intermediate gaps between two known data points, and carry-forward of the last value (held constant) to 2050. Missing routing pathways were filled programmatically based on the known system structure, linking each collection stream to chemistry-specific sorting and treatment nodes. From the treatment node, chemistry-specific sorting routes were filled, with each battery chemistry assigned to its corresponding sorting pathway (see Fig. 1).

After gap filling, a mass balance check was performed across all nodes to ensure physical consistency. Each TC was then assigned to its respective scenario, reference year, and the observed (OBS) data category where applicable. As a last step, columns were restructured and renamed to conform to the input format required by the downstream recovery model.

2.7. Back-calculation of scenario-specific transfer coefficients

Building on the scenario narratives defined in Section 2.5, this section describes how the corresponding step-level TCs were derived from the regulatory recovery targets. For a given recycling route, the overall recovery of element I equals the product of the step-level TCs across all n sequential process steps:

$$R_{route,i} = \prod_{j=1}^n TC_{i,j} \quad (3)$$

Since a given target can be met by multiple step-level combinations, a feasibility-guided back-calculation is applied rather than numerical optimisation. For each route and scenario, process steps were classified as fixed or adjustable by technical improvement potential: near their practical limit (pyrometallurgical smelting) were held at observed values, while steps with realistic headroom (mechanical pre-treatment, hydrometallurgy, hydrometallurgical slag-treatment) were increased until the product met the respective regulatory target (see Supplementary Information). Where a target could not be reached by downstream alone, otherwise fixed steps were also increased. For routes exporting black mass, retained TCs were set such that the target remains attainable once downstream recovery is applied. All adjusted TCs remain within technically plausible bounds ($TC \leq 1$) and non-decreasing over time. The steps adjusted per route and scenario are summarised in Table 2.

Table 2

Feasibility-guided adjustment of TCs per LIB recycling route 1 to 4 and scenario (BAU, REC). For each route, process steps were classified as *fixed* (held at observed values) or *adjusted* (increased until the end-to-end recovery met the respective regulatory target), based on their technical improvement potential.

Route	Scenario	Fixed and adjusted steps	Rationale
1	BAU	Pyrometallurgy (fixed), hydrometallurgy (Co, Cu), and slag hydrometallurgy (Li)	Pyrometallurgical TCs hard to improve; TCs for Li only via new slag treatment
	REC	Pyrometallurgy (Co, Cu) and hydrometallurgy (Co, Cu, Ni), and slag hydrometallurgy (Li, Fe, Mn, P, Al)	REC targets too high to reach without also raising TCs for pyrometallurgy; improved slag treatment for various metals
2	BAU	Mechanical treatment (Co, Cu, Li, Fe, Mn, P, Al, Graphite) and hydrometallurgy (Co, Ni, Graphite)	Both steps have realistic headroom; graphite is assumed recoverable via hydrometallurgy since no pyrometallurgical step is present and it is therefore not combusted/lost
	REC	Mechanical treatment (Co, Cu, Li, Fe, Mn, P, Al, Graphite) and hydrometallurgy (Co, Ni, Fe, Mn, P, Al, Graphite)	Same steps as BAU, higher target levels
3	BAU/ REC	Mechanical treatment (Co, Cu, Li, Fe, Mn, P, Al, Graphite)	Only mechanical step present; TCs must be set higher so that regulatory targets can still be met through the downstream recovery processes applied to the exported black mass
			Pyrometallurgy hard to improve; TCs for Li only via new slag treatment
4	BAU	Pyrometallurgy (fixed), mechanical treatment (Co, Cu, Li, Fe, Mn, P, Al), hydrometallurgy (Co, Ni), and slag hydrometallurgy (Li)	
	REC	Mechanical treatment (Co, Cu, Li, Fe, Mn, P, Al), pyrometallurgy (Co, Ni), hydrometallurgy (Co, Ni) and added slag hydrometallurgy (Li, Fe, Mn, P, Al)	REC targets too high to reach without also raising TCs for pyrometallurgy; improved slag treatment for various metals

2.8. Heuristic uncertainty classification and deterministic propagation

Given the hierarchical structure of the recovery model, in which the end-to-end recovery of a route is the product of its sequential step-level TCs (Section 2.4), uncertainties in individual coefficients propagate multiplicatively and accumulate along the process chain. To assess this quantitatively, each TC was assigned an assumption-based uncertainty range from a two-dimensional quality classification considering the maturity of the underlying data and the temporal distance of the projection.

The first dimension, data quality, follows the source classification of Table 1 (I/P/L-scale as well as assumption/review-based values). The second, temporal quality, reflects the increasing uncertainty of longer-term projections graded by period (2010–2025, 2026–2030, 2031–2040, 2041–2050). Each TC is located within a classification matrix (Table 3), mapping data and temporal quality to an uncertainty range of $\pm 5\%$ to $\pm 20\%$. The values of $\pm 5\%$, $\pm 10\%$, $\pm 15\%$, and $\pm 20\%$ are not statistically derived, as most sources report single point values without dispersion measures. They are assumed, heuristic bounds based on expert judgement, with the narrowest band assigned to verified industrial-scale mass balances for near-term periods and the widest to assumption- or review-based values extrapolated to 2050, increasing in equidistant steps to keep the scheme transparent. For conventional processes whose TCs are held constant over time, the temporal

Table 3

Heuristic uncertainty classification matrix combining data quality and temporal quality. Each cell gives the assumed uncertainty range ($\pm$ %) applied to the corresponding TC (values reflect expert judgement on source reliability and are not statistically estimated confidence intervals).

Temporal Quality	2010–2025	2026–2030	2031–2040	2041–2050
Data Quality	1	2	3	4
Industrial (I)	1 ± 5 %	± 10 %	± 10 %	± 15 %
Pilot (P)	2 ± 10 %	± 10 %	± 15 %	± 15 %
Lab (L)	3 ± 10 %	± 15 %	± 15 %	± 20 %
Assumptions/ Review Paper	4 ± 15 %	± 15 %	± 20 %	± 20 %

dimension is fixed to the earliest period.

The step-level uncertainty ranges were propagated through the multiplicative route structure by combining the respective lower and upper bounds of all steps (subject to $TC \leq 1$), yielding a minimum and maximum recovery rate per route, scenario, and reference year. Since the procedure propagates lower and upper bounds rather than a probability distribution, the resulting intervals are deterministic sensitivity bounds delimiting a plausible envelope, not confidence intervals. Full details on uncertainty classification, assigned ranges per TC, and resulting minimum and maximum recovery rates are documented in the [Supplementary Information](#).

2.9. Structural decomposition of system-level recovery

To disentangle the two mechanisms driving changes in system-level recovery, which are shifts in the route composition versus improvements in process efficiency, a structural decomposition was applied. For selected elements, the system-level recovery R_{sys} is expressed as the share-weighted sum of route-specific technical recovery rates:

$$R_{sys} = \sum_r s_r R_r \quad (4)$$

The parameter s_r denotes the share of route r and R_r its end-to-end technical recovery. The change in R_{sys} between baseline (OBS, 2024) and scenario endpoint (BAU or REC, 2050) is decomposed into a *composition effect* and an *efficiency effect* using a symmetric (Shapley-consistent) mean-value decomposition:

$$\Delta R_{sys} = \underbrace{\sum_r (s_r^1 - s_r^0) \frac{R_r^0 + R_r^1}{2}}_{\text{composition effect}} + \underbrace{\sum_r \frac{s_r^0 + s_r^1}{2} (R_r^1 - R_r^0)}_{\text{efficiency effect}} \quad (5)$$

The composition effect isolates route-share reallocation at constant recovery performance, whereas the efficiency effect captures route-internal TC improvements at constant route shares. The symmetric formulation ensures the two effects sum exactly to the observed net change, leaving no interaction residual. The decomposition was performed for each regulated and non-regulated element and both scenarios and can be found in the [Supplementary Information](#).

3. Results

3.1. Scenario-based distribution of LIB recycling routes

The following LIB route shares represent scenario-specific capacity allocations that determine the weighted TCs entering the MFA model. The five recycling routes are applied in varying proportions over time ([Fig. 3](#)). The data indicate a marked shift from purely mechanical treatment (Route 3) to more integrated multi-step approaches (Routes 1, 2, and 4), while Route 5 emerges from 2032 onward. In 2024, mechanical treatment (Route 3) accounts for nearly 80 % of the total treatment share in both scenarios, reflecting its current dominance in the EU.

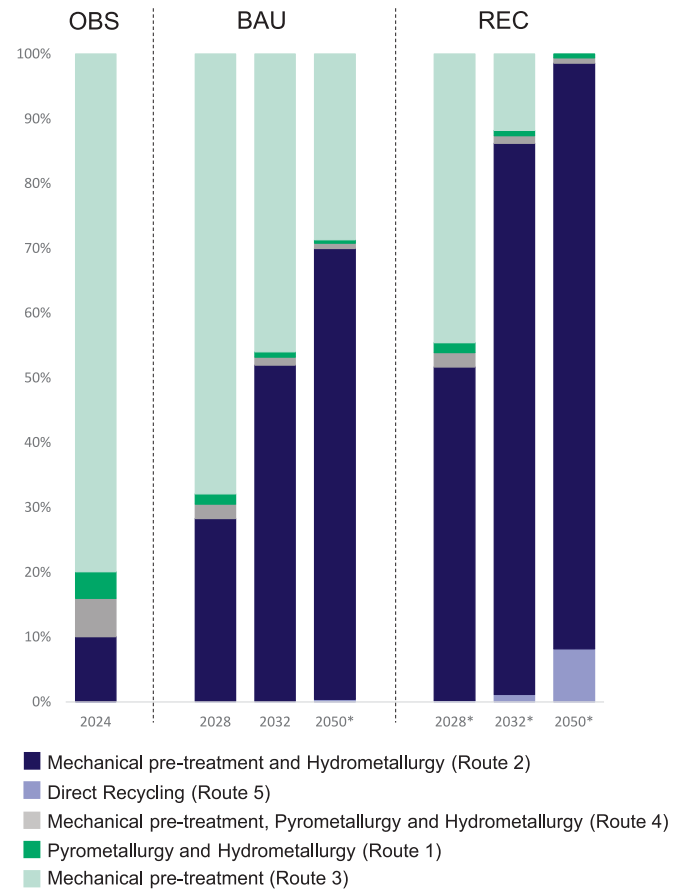


Fig. 3. Projected share of LIB recycling routes in Europe, 2024–2050, based on announced plant capacities for BAU (2028, 2032) and estimated technology trajectories for REC (2028*, 2032* and 2050*).

Under BAU, Route 3 declines to 29 % by 2050, as black mass from mechanical treatment is currently exported from the EU for hydrometallurgical processing (80 to 90 %), with this share declining as domestic capacities are established. Route 2 grows from 10 % in 2024 to 70 % by 2040, becoming the dominant route. Routes 1 and 4 decline, reflecting higher energy demand and lower recovery efficiencies associated with pyrometallurgical steps, and Route 5 reaches 0.4 % by 2040.

The REC scenario assumes a more rapid transition. Route 3 drops to zero by 2050, reflecting the assumption that sufficient domestic hydrometallurgical capacity eliminates black mass export. Route 2 reaches 90 % by 2050, reflecting expected upscaling of integrated recycling systems. Route 5 grows substantially to 8 % by 2050.

3.2. Scenario-specific recovery rates by route and element

TCs for the individual recycling routes were derived in alignment with EU Battery Regulation recovery targets ([Section 1.3](#)) for the regulated elements Co, Li, Ni, Cu, and Pb. For all other materials (Fe, Mn, phosphorous (P), aluminium (Al), graphite, ferrous metals), no regulatory targets exist, and their recovery results from assumed step- and route-specific TCs. Recovery rates were differentiated by battery component, as these undergo distinct treatment steps (e.g. Fe from the casing is treated differently from Fe in LFP cathode active material (CAM)).

The comparative recovery analysis in the main text focuses on Routes 1, 2, and 4, as these represent integrated routes that recover target materials within the modelled EU system boundary and can therefore be benchmarked against the regulatory targets on a technical-recovery basis ([Section 2.4](#)). Route 3 is excluded because its black mass is an

intermediate fraction under Regulation (EU) 2025/606 that is largely exported and thus does not yield creditable output. Route 5 is excluded as a pilot-scale, LFP-specific route whose constant TCs are not target-driven. For completeness, all five routes are reported in the [Supplementary Information](#) (Table S1 and S2). For regulated elements, BAU and REC values reflect TC improvements necessary to meet regulatory targets. For non-regulated materials, values emerge from modelled process configurations without imposed targets. The OBS

2024 baseline shows the lowest recovery rates (e.g. Route 4: 0 % Li and 64 % Co). These route- and element-specific values are analysed in [Section 4.1](#).

3.3. Transfer coefficients of battery recycling technologies

The computation of TCs was performed using a Python script, yielding a dataset in CSV format comprising 36,358 individual TC

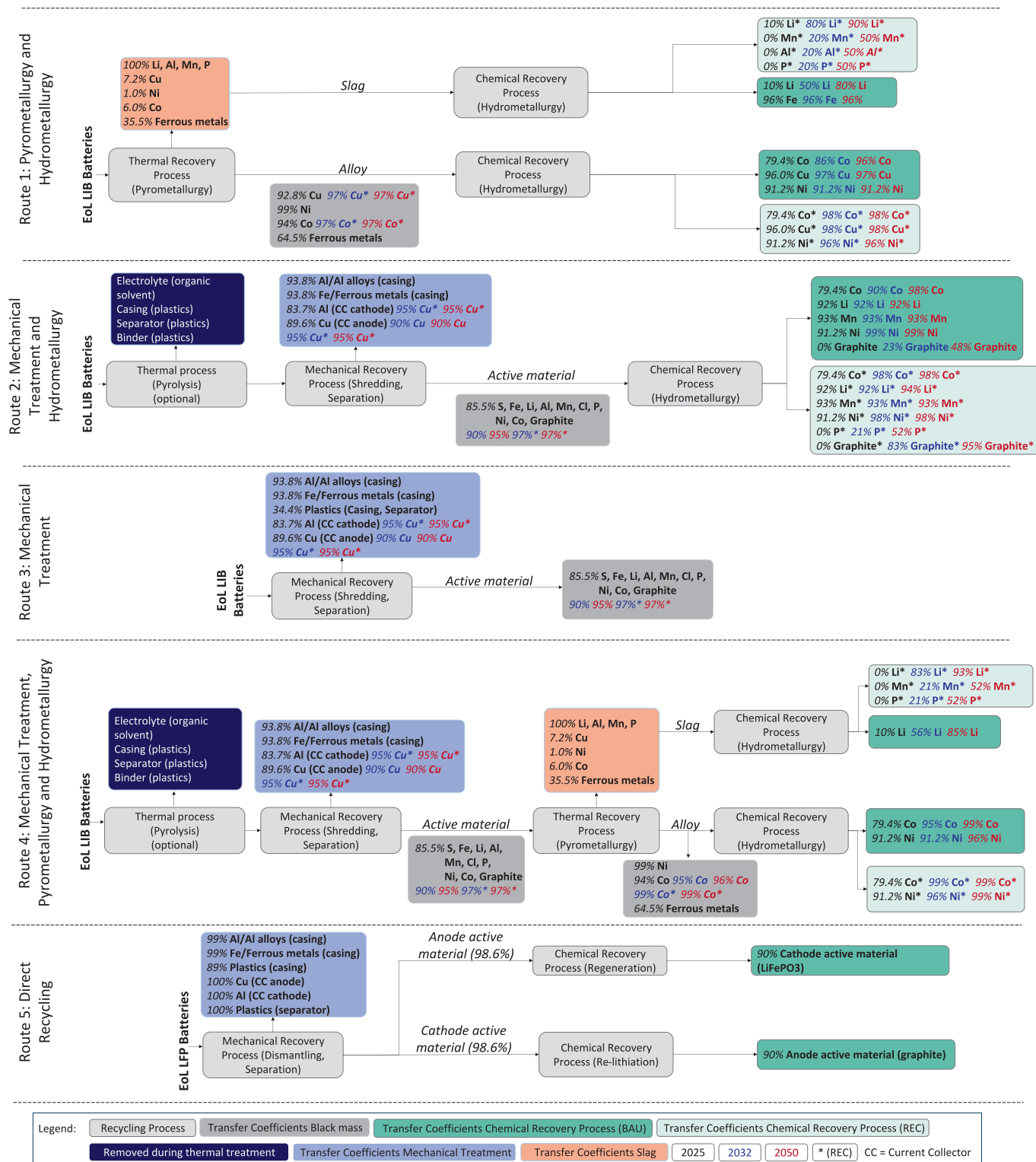


Fig. 4. Transfer coefficients of LIB recycling technologies.

values. The dataset is publicly available on Zenodo (Maisel et al., 2026a). TCs are provided from 2010 onwards annually and incorporate all scenarios from 2024 to 2050.

3.3.1. Transfer coefficients of LIB recycling technologies

Route 1 is one of the most established industrial LIB recycling approaches, operated commercially by Umicore (Fig. 4). TCs for the pyrometallurgical step are taken from the Umicore patent (Cheret and Santen 2007) and applied as proxy values for all pyrometallurgical steps due to data constraints. LIB cells are fed into a high-temperature furnace without pre-treatment, yielding a Co-Ni-Cu-Fe alloy and a slag containing Li, Al, and Mn (assumed fully retained). Graphite and electrolyte are combusted and not recovered. The alloy is refined hydrometallurgically into battery-grade metals. Under BAU, TCs for target metals remain moderate with gradual Li improvement. Under REC, TCs for Co, Cu, and Li increase substantially, and Mn, Al, and P are partially recovered.

Route 2 combines pyrolysis, mechanical pre-processing, and hydrometallurgical recovery. Pyrolysis decomposes organic components (electrolytes, binders, solvents), which are assumed to be fully lost. Mechanical processing then generates black mass and recovers casing, ferrous, and current-collector fractions. Black mass is subsequently processed hydrometallurgically. Under BAU, hydrometallurgical TCs improve moderately, with gradual graphite recovery development.

Under REC, black mass recovery increases to 97 %, TCs for Co, Ni, and Li reach higher levels, graphite recovery increases substantially, and P is partially recovered.

Route 3 comprises mechanical treatment only, generating black mass. Under the methodology of Regulation (EU) 2025/606, black mass is an intermediate fraction credited only after subsequent downstream processing. As this black mass is largely exported (estimated 80–90 %) in current EU practice, it represents a domestic loss and a non-creditable output at the point of export, despite possible technical and creditable recovery abroad. (see Section 3.2). Under BAU, recovery improves moderately, while under REC, it reaches 97 %.

Route 4 integrates pyrolysis, mechanical separation, pyrometallurgical processing, and hydrometallurgical recovery. Pre-treatment follows Route 2, while the resulting black mass is processed pyrometallurgically to produce a metal alloy, with non-target elements (Li, Al, Mn, P) transferred to slag. Hydrometallurgy is applied to both alloy and slag. Under BAU, TCs for Co, Ni, and Cu are high, while Li recovery via hydrometallurgical slag treatment starts at zero and improves gradually. Under REC, black mass recovery and TCs for Co and Ni reach up to 99 %, Li recovery increases substantially, and Mn, Al, and P are partially recovered.

Route 5 is a direct recycling approach for LFP chemistries, aiming to preserve and restore cathode and anode active material (AAM) functionality via re-lithiation and graphite regeneration. As direct recycling

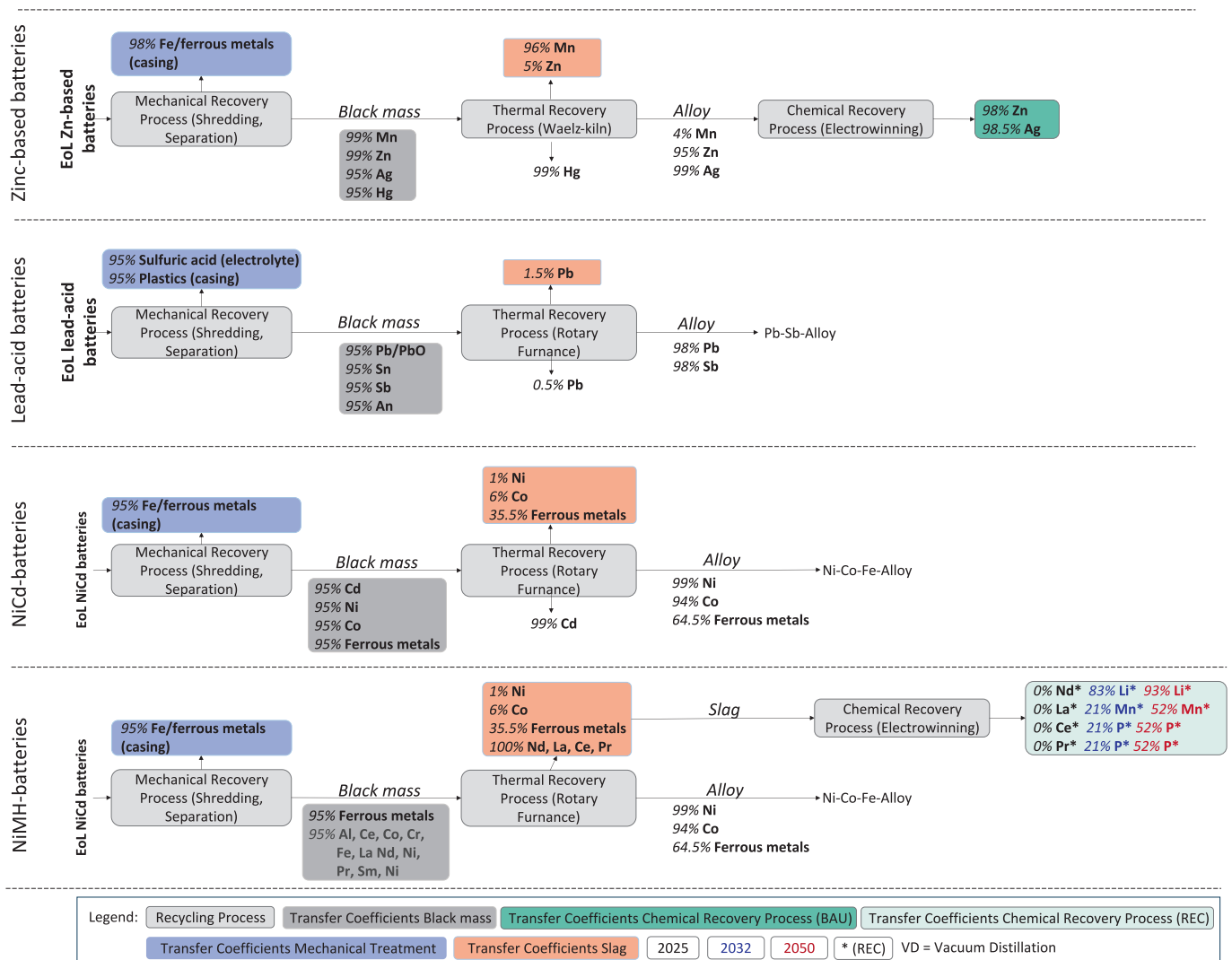


Fig. 5. Transfer coefficients of conventional battery recycling technologies.

is at pilot scale, values remain constant across scenarios.

3.3.2. Transfer coefficients of conventional battery recycling technologies

For each conventional battery type, a representative industrial process is assumed, reflecting commonly applied practices. TCs are held constant across scenarios, except for rare earth element (REE) recovery from NiMH under REC scenario. Specific values are shown in Fig. 5. Zinc-based battery recycling involves mechanical treatment, thermal recovery via Waelz kiln, and electrowinning. Mechanical treatment separates an Fe fraction and black mass. The latter is processed in the Waelz kiln to yield Zn, silver (Ag), and mercury, with residual manganese (Mn) and Zn in slag, followed by electrowinning to purify commercial-grade Zn and Ag.

Lead-acid battery recycling is well-established. It comprises crushing, separation, smelting in a rotary furnace, and refining. Mechanical processing recovers a Pb-bearing black mass containing Pb, PbO, antimony (Sb), and tin (Sn) and non-metallic fractions at ~ 95 %. The black mass is smelted into a Pb-Sb alloy with high recovery, with minor Pb dust and slag losses.

NiCd batteries are largely phased out (consumer sales banned in Europe in 2017), but recycling of legacy waste remains essential. Mechanical treatment yields a black mass containing Cd, Ni, Co, and ferrous metals at 95 % recovery. Black mass is treated by vacuum distillation, separating Cd at 99 % purity, followed by pyrometallurgical processing into a Ni-Co-Fe alloy (Zhan and Xu 2014).

NiMH battery recycling resembles NiCd but omits vacuum distillation (no Cd). Mechanical treatment yields a black mass containing Ni, Al, Fe, Co, and REE (cerium (Ce), chromium (Cr), lanthanum (La), neodymium (Nd), praseodymium (Pr), and samarium (Sm)), followed by pyrometallurgical processing. REEs are fully retained in slag and not recovered under BAU (Salehi et al., 2023). Under REC, an additional hydrometallurgical step enables REE recovery at an estimated efficiency of 60 % in 2050.

3.4. Drivers of system-level recovery: composition versus efficiency

To clarify whether system-level recovery gains are driven by route reallocation or by efficiency improvements, a structural decomposition (Section 2.9) was performed for six representative elements. Fig. 6

presents composition and efficiency effects for BAU and REC scenarios.

Across almost all elements, the efficiency effect is the dominant positive driver and intensifies from BAU to REC, reflecting the more ambitious target-compliant TC trajectories, e.g., graphite + 23.2 to + 49.9 percentage points (pp) and copper + 0.4 to + 5.2 pp. In contrast, the composition effect is negative for cobalt, lithium, nickel, and graphite and becomes more negative under REC, e.g., cobalt – 6.0 to – 14.4 pp. This counterintuitive result reflects the system boundary: Route 3 generates black mass that leaves the modelled boundary as an intermediate fraction, counting towards technical recovery at the boundary but not towards domestic secondary supply. Aluminium is the sole exception, with a dominant positive composition effect in both scenarios (+3.3 and + 2.8 pp).

Importantly, modest net changes can conceal large opposing effects. For cobalt, a net gain of + 5.5 pp under REC masks a composition effect of – 14.4 pp and an efficiency effect of + 19.9 pp. For graphite, the net change even reverses sign, declining by 9.4 pp under BAU but increasing by 20.5 pp under REC. Process efficiency, rather than route composition, therefore emerges as the primary lever, with route composition acts as a secondary and frequently counteractive effect whose direction is strongly element-specific.

3.5. Use of transfer coefficients in material flow analysis

The derived TCs can be integrated into MFA by applying them sequentially at each process node. Starting from EoL battery input, outgoing flows at each node are calculated by multiplying the incoming mass by the respective TC. To illustrate this approach, Fig. 7 presents an exemplary Sankey diagram for the Li recycling chain. Flow widths result from the successive application of TCs, and end-node percentages give each output's share of the initial input, making transparent where material is retained and where losses occur.

This is further operationalised on the Urban Mine Platform (urbanmineplatform.eu), where battery material flows are linked with the derived TCs. The platform provides additional Sankey diagrams for various material streams and scenarios, enabling users to interactively explore how process efficiencies along the recycling chain affect overall material recovery at EU scale.

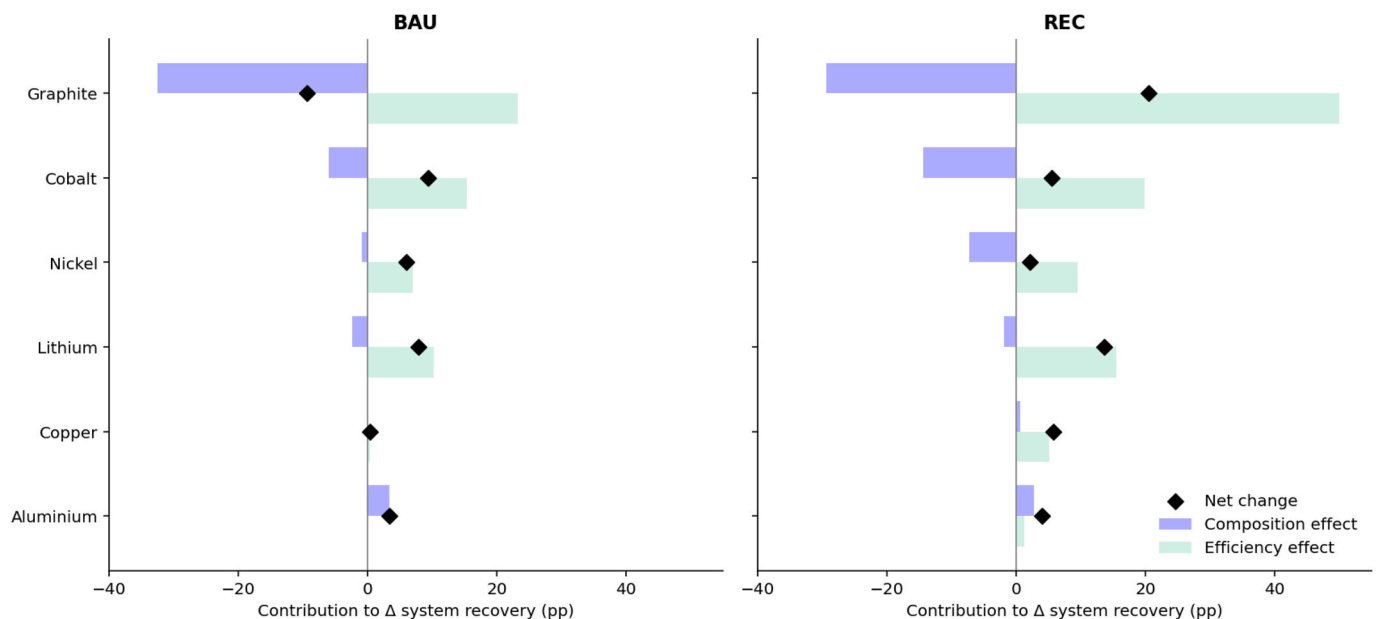


Fig. 6. Structural decomposition of the change in system-level recovery (2024 to 2050) into composition and efficiency effects for the BAU (left) and REC (right) scenarios. Diamonds indicate the net change (ΔR_{sys}).

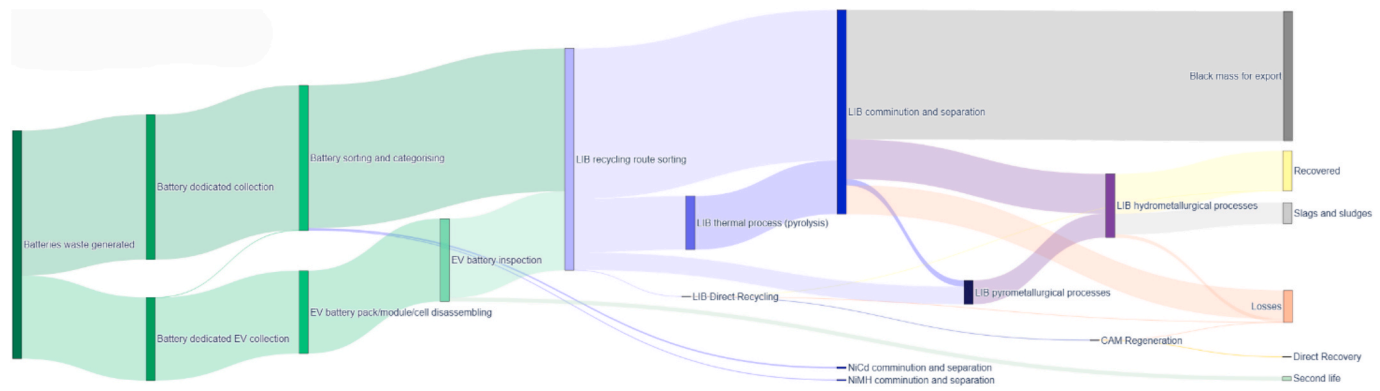


Fig. 7. Sankey diagram of the battery recycling chain exemplarily for lithium.

4. Discussion

4.1. Gap Analysis: Current performance vs. Regulatory targets

The gap analysis is conducted for the three integrated routes (1, 2, 4) that generate target-material output within the modelled EU system boundary. The analysis is conducted on a technical-recovery basis (Section 2.4). Routes 3 and 5 are discussed separately (Sections 3.2 and 4.2) and are not evaluated against the regulatory targets on the same basis. The EU Battery Regulation establishes minimum recovery rates for Co, Ni, Cu, and Li that must be met by 31 December 2027 and 31 December 2031, while recovery is reported for the reference years 2028 and 2032 (Section 1.3). A comparison of OBS recovery rates in 2024 with these targets reveals substantial and unevenly distributed gaps across elements and routes. For Co, the OBS baseline ranges from 64 % (Route 4) to 74 % (Route 1), translating into a gap of 16 to 26 % relative to the target of 90 % due by 31 December 2027. Under the BAU scenario, only Route 2 is assumed to reach this target by end of 2031, while Routes 1 and 4 require the full modelled period to approach 90 to 95 %. For Ni, the gap is comparatively smaller: OBS values range from 77 % (Route 4) to 90 % (Route 1), placing Route 1 already at the target level required by the 2027 deadline. Cu presents the smallest gap among regulated elements, with OBS values of 89 to 90 % across all routes, effectively meeting the regulatory target under current practice.

In contrast, lithium represents the most critical gap. OBS recovery ranges from 0 % (Route 4) to 78 % (Route 2), revealing fundamentally different starting positions depending on route configuration. Route 1 achieves only 10 % in the baseline, and Route 4 currently recovers no lithium at all, as slag treatment has not yet been implemented at scale. Meeting the regulatory target of 50 % by end of 2027 thus requires a gap closure of 40 to 50 percentage points for Routes 1 and 4 within less than five years. However, the Regulation defines uniform recovery thresholds without differentiating by recycling route, despite the finding that route-specific feasibility varies by up to 78 percentage points for Li alone. No industrial-scale precedent exists for the implied improvement trajectory, raising fundamental questions about the adequacy of the regulatory timeline. Route 2, with an OBS value of 78 %, already exceeds the 50 % requirement (2027 deadline) but remains below the target of 80 % (2031 deadline).

Beyond regulated elements, Fe, Mn, P, Al from CAM, and graphite show zero recovery across most routes in the OBS baseline. These materials are not subject to regulatory targets, and their recovery depends entirely on deliberate technology development and economic incentives. The Regulation provides no direct incentive for their recovery, creating a structural blind spot for materials that are nonetheless relevant for circularity and supply security. At the route level, the gap analysis reveals structural differences in target achievability. Route 1 shows high baseline performance for Co and Ni but is fundamentally limited in recovering materials beyond the metal alloy. Route 2

demonstrates the most balanced profile, with relatively high OBS values across regulated elements and meaningful graphite recovery. Route 4 combines the limitations of pyrometallurgy with the potential of hydrometallurgical slag treatment, resulting in the widest gap range across elements.

When interpreting these gaps, the three recovery perspectives introduced in Section 2.4 must be distinguished. The recovery rates reported here represent technical recovery (the cumulative product of step-specific TCs) and do not directly correspond to the regulatory-credited recovery under Regulation (EU) 2025/606. In particular, lithium retained in slag (Routes 1 and 4) is technically present but is explicitly excluded from regulatory-credited recovery whereas lithium extracted from the slag by subsequent hydrometallurgical treatment is creditable at the RM calculation point. Black mass generated in Route 3 is an intermediate fraction whose contained metals become creditable only after downstream processing. Consequently, the exported black mass under BAU (Section 3.1) constitutes a loss of domestic secondary material from the EU perspective, even where the metals may still be technically recovered and regulatorily credited abroad, provided equivalent conditions are documented (Art. 72(3) of Regulation (EU) 2023/1542). The gaps reported here should therefore be read as technical-recovery gaps. The regulatory compliance gap is systematically at least as large, and for Routes 1 and 4 larger, because slag-borne lithium is excluded from creditable output (Section 1(4) of the Annex to Regulation (EU) 2025/606).“.

4.2. Composition versus efficiency in system-level recovery

The structural decomposition (Section 3.4) qualifies a conclusion that is easily misread from aggregate scenario outcomes. Although the transition towards integrated hydrometallurgical routes is often framed as the principal driver of improved recovery, the decomposition shows that process-level efficiency dominates the system-wide gains for nearly all elements, whereas route reallocation is secondary and, for cobalt, nickel, lithium, and graphite, even counteractive. This apparent paradox is a direct consequence of the system boundary adopted. Black mass from mechanical-only treatment (Route 3) leaves the modelled boundary as an intermediate fraction, so its contained metals are attributed to Route 3 as technical recovery at the boundary but do not contribute to domestic secondary supply. Its phase-out therefore lowers measured system-level recovery unless offset by efficiency improvements in the routes that replace it. The negative composition effect is thus partly a boundary-attribution effect rather than a physical loss, and it underscores that system-level recovery figures are sensitive to where the boundary is drawn and how exported intermediate streams are attributed.

In our analysis, is counted as technical recovery at the point of black-mass generation, but not as domestic secondary material. Under the regulatory metric it is likewise not creditable at the point of export, since

Section 1(5) of the Annex to Regulation (EU) 2025/606 classifies black mass as an intermediate fraction that cannot be considered an output fraction, so that its contained metals count towards recovery only after verified downstream processing at the RM-calculation point (European Commission, 2023, 2025). Where aggregate recovery indicators do attribute exported black mass to recovery at the system boundary, they may overstate the circularity actually achieved within the EU. In the case of graphite, where the net change reverses sign between scenarios, demonstrates that net recovery outcomes cannot be interpreted without decomposition, as they may conflate large, opposing structural and efficiency effects. For policy evaluation, this suggests that recovery targets should be complemented by boundary-consistent, step-specific reporting so that genuine efficiency gains can be distinguished from shifts in treatment structure.

4.3. Limitations and uncertainties

4.3.1. Data Availability, Representativeness and Proxy Assumptions

The derivation of TCs relies predominantly on published literature, patents, and industry reports. For some process steps, particularly hydrometallurgical slag treatment and graphite recovery, only limited empirical data are available, often from laboratory- or pilot-scale experiments rather than commercial operations and their transferability to industrial practice remains uncertain. The literature reports recovery efficiencies under varying experimental conditions (e.g., acid concentration, temperature, solid-to-liquid ratio), which are not always fully documented. To make this heterogeneity transparent, each reference in Table 1 is classified by its scale of origin (L/P/I/R), serving as a qualitative confidence indicator: industrial-scale (I) values are considered most representative, whereas lab- (L) and pilot-scale (P) values carry higher uncertainty regarding their transferability to full-scale operation.

More fundamentally, most recycling facilities provide little to no transparency regarding process-level recovery performance, which is particularly pronounced for emerging LIB recycling technologies where multiple process combinations are developed simultaneously. Where only individual process steps data was available, they were extrapolated to address data gaps and, in several cases, transferred to combined recycling routes, introducing further uncertainty. The TCs must therefore be interpreted with caution, as they are subject to high variability and based on estimates or expert assumptions. Improved transparency from recycling operators, particularly through consistent and verifiable reporting aligned with regulatory recovery targets, would substantially strengthen the empirical basis for future TC derivation.

For NiMH, NiCd, lead-acid, and zinc-based batteries, the assumption of constant TCs reflects their technological maturity of these processes but precludes the possibility of future efficiency losses due to ageing infrastructure or shifting economic viability reducing actual recovery performance below the assumed levels.

Due to data availability constraints, TCs from the Umicore pyrometallurgical patent (Cheret and Santen 2007) were adopted as proxy values for all pyrometallurgical steps across Routes 1 and 4. While this patent provides one of the most detailed publicly available mass balances for industrial pyrometallurgy, other operators may employ different furnace configurations, temperature profiles, or feed compositions that yield different element distributions between alloy and slag. The assumption of full retention of Li, Al, and Mn in the slag represents a further simplification, as partial losses to off-gas, dust, or other phases may occur in practice (Vest et al., 2010).

4.3.2. Scenario assumptions and route allocation uncertainty

The route shares and TC improvements assumed under the BAU and REC scenarios are not predictions but explorative boundary conditions. The 2028 and 2032 route shares are grounded in announced plant capacities, which carry their own uncertainty (e.g., delays in commissioning, changes in business strategy). The 2050 route distribution is estimated based on expected technology trajectories and is therefore

subject to considerable uncertainty. Similarly, assumed TC improvements are modelled as linear or stepwise trajectories, whereas real-world technology development may follow non-linear patterns driven by discrete innovation events, policy changes, or market shifts.

Several further uncertainties affect the route allocation. Announced capacities do not necessarily translate into operational capacity, due to commissioning delays, strategy changes, or feedstock shortages. The fast-moving LIB recycling market is difficult to capture at any single point in time (Wagner-Wenz et al., 2023). Assigning individual facilities to one of the five defined routes involves simplification, as commercial processes often combine elements from multiple routes or employ proprietary variations. Wagner-Wenz et al., (2023) demonstrate that LIB recycling typically involves between 2 and 14 process steps per route, giving rise to manifold combinations that resist simple categorisation. The route distribution for 2050 is entirely assumption-based, as no reliable capacity announcements exist for this time horizon. The assumed continued dominance of hydrometallurgical approaches (Routes 2 and 4) reflects current investment trends but does not account for potential disruptive developments, such as rapid maturation of direct recycling or novel process concepts. Because route shares based on announced capacities inherently favour mature technologies (pyrometallurgy at TRL 9 versus direct recycling at TRL 4), the long-term potential of emerging routes may be underestimated (Wagner-Wenz et al., 2023). The modelled route distribution should therefore be interpreted as a plausible but uncertain allocation rather than a definitive market forecast.

4.3.3. Chemistry-specific limitations and mass balance closure

As noted in Section 4.1, the input material composition can influence the TCs of recycling processes. Wilke et al., (2024) demonstrate that input composition, process configuration, and technological maturity strongly influence TCs and, consequently, calculated recycling efficiencies and recovery potentials (Wilke et al., 2024). Using case studies from battery recycling, the authors show that changes in input flows can directly alter TCs, challenging the assumption of static process parameters. Individual TCs therefore represent only a snapshot corresponding to a specific and well-defined input stream. This applies in particular to LIBs, whose recycling is chemically and technologically heterogeneous and by processes that are still evolving. In contrast, well-established and largely standardised recycling processes, such as lead-acid battery recycling, exhibit significantly lower uncertainty (Wilke et al., 2024).

In this study, the TCs for LIB recycling are defined for a generic LIB input stream without differentiation by cathode chemistry, except for direct recycling (Route 5) explicitly designed for LFP batteries. This represents an important limitation, since different cathode chemistries can yield systematically different recovery pathways. The literature indicates that Co/Ni-free chemistries (e.g., LFP) generally perform worse in pyrometallurgical routes optimised for high-value metals such as Co and Ni, while lithium is typically transferred to the slag phase (Latini et al., 2022). Consequently, applying uniform TCs across all LIB chemistries may overestimate recovery efficiencies for Co/Ni-poor batteries and underestimate chemistry-specific losses.

Conversely, recent studies suggest that the role of LFP in recycling systems is more nuanced. Zou et al., (2024) highlight that mixed-chemistry recycling can, under certain conditions, be advantageous, showing that LFP may act not only as a reductant in pyrometallurgical processes but also as a source of phosphate ions that can facilitate downstream impurity removal, particularly for Fe and Al (Zou et al., 2024). This indicates that interactions between chemistries in mixed input streams can alter process behaviour and recovery outcomes, potentially affecting TCs in ways that are not captured by chemistry-agnostic assumptions (Zou et al., 2024).

Finally, the constraint that all TCs for a given element must sum to unity (Equation (2)) imposes internal consistency but means that overestimation of recovery into one output stream understates losses

elsewhere. Where individual TCs were back-calculated from reported recovery rates without a fully documented mass balance, the allocation of residual fractions to loss streams introduces additional uncertainty.

4.4. Theoretical, practical and policy implications

The findings carry policy, practical, and theoretical implications. From a policy perspective, three points follow directly from the gap analysis (Section 4.1). First, Co, Ni, and Cu targets appear achievable under current technology trajectories, suggesting that policy attention should prioritise Li and graphite recovery as the binding constraints. Second, meeting the Li recovery requirement due by 31 December 2027 for Routes 1 and 4 requires rapid deployment of slag treatment capacity for which industrial readiness is not yet demonstrated, calling into question whether the regulatory timeline is grounded in technological reality. Third, the absence of targets for non-regulated materials implies that their contribution to secondary supply will remain negligible unless additional policy instruments are introduced. Beyond these, the decomposition results (Section 4.2) suggest that recovery targets should be complemented by boundary-consistent, step-specific reporting, so that genuine efficiency gains can be distinguished from shifts in treatment structure.

From a practical perspective, the compiled route-specific TC dataset provides recyclers and MFA practitioners with a directly usable parameter base, while the route-specific results identify industrial-scale slag treatment for lithium and domestic black-mass processing as the highest-leverage intervention points. Importantly, both the TC dataset and the underlying Python-based recovery model are publicly available, making the analysis fully reproducible and allowing users to update individual coefficients as new industrial-scale data become available or as process technologies evolve. The framework can thus be maintained as a living parameter base rather than a static snapshot.

From a theoretical and methodological perspective, the study shows that multiplicatively linked step-level TCs bridge process- and system-level recovery, whereby a single low coefficient can constrain end-to-end recovery. The structural decomposition further demonstrates that aggregate indicators can conceal or even reverse opposing composition and efficiency effects (as observed for graphite), establishing boundary-consistent, decomposition-based TC accounting as a transferable framework for recycling MFA beyond the battery domain.

5. Conclusion

This study proposes a route-specific set of TCs for established and emerging battery recycling routes, intended to approximate future secondary raw material availability from EoL batteries in material flow models. By compiling available empirical data into 36,358 scenario-ready TC values, the work offers a structured, albeit uncertain, basis for bridging process-level recycling evidence and system-level secondary supply estimation for zinc-based, lead-acid, NiCd, NiMH, and LIBs.

The scenario-based analysis suggests that the gap between current recycling performance and potential recovery is highly element-specific. While estimated TCs for Co, Ni, and Cu already approach regulatory thresholds, Li recovery from pyrometallurgy-based routes remains near zero in technical-recovery terms, and is non-creditable where lithium remains in the slag. For non-regulated materials such as Fe, Mn, P, Al from CAM, and graphite, TCs suggest zero or marginal recovery, implying that their contribution to secondary supply depends entirely on deliberate technology development. A structural decomposition demonstrates that process-level efficiency is the dominant driver of system-wide recovery for most elements, whereas route composition acts as a secondary and often counteractive effect.

However, the proposed TC dataset is subject to considerable uncertainty. Values for emerging steps rely on pilot-scale data, chemistry-agnostic TCs do not capture cathode chemistry differences, and extrapolation to 2050 is speculative. Results should be interpreted as

indicative approximations rather than definitive projections.

The analysis raises fundamental questions about the EU Battery Regulation's target architecture. The Regulation defines uniform recovery thresholds without differentiating by recycling route or cathode chemistry, despite route-specific feasibility varying by up to 78 % for lithium alone. Meeting the 50 % Li requirement by 31 December 2027 requires a gap closure of 40 to 50 % for pyrometallurgy-based routes within less than five years, without industrial-scale precedent. Furthermore, by covering only Co, Ni, Cu, Li, and Pb, the Regulation provides no incentive for recovering graphite, manganese, or phosphorus, creating a structural blind spot and may undermine broader circularity objectives, while the absence of mandatory step-specific reporting limits compliance verification.

CRediT authorship contribution statement

Franziska Maisel: Writing – original draft, Visualization, Methodology, Investigation, Formal analysis, Conceptualization. **Max Tippner:** Writing – review & editing. **Kibria Mainuddin:** Software. **Tales Yamamoto:** Writing – review & editing. **Giulia Iattoni:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the authors used Claude Opus 4.8 R for language editing and refining grammar, spelling and sentence structure. The authors reviewed and edited the output as needed and take full responsibility for the content of the published article.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.wasman.2026.115873>.

Data availability

I have shared the link to my data and code in the submission.

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